

Homocyst(e)ine

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Screen patients who may be at risk for heart disease and stroke

Limitations

This test is not intended for use in the diagnosis of folate or vitamin B₁₂ deficiency.

Custom Additional Information

Severe homocysteinemia is typically caused by a rare inborn error of metabolism.^{1,2} The most common defect that can produce levels >100 $\mu\text{mol/L}$ is homozygous cystathionine- β -synthase (CS) deficiency, which occurs with an incidence of 1 per 300,000 live births. About 1% of the population has heterozygous CS deficiency, a condition that typically results in moderate to intermediate hyperhomocysteinemia. Individuals with CS deficiency are at increased risk for occlusive vascular disease.^{1,2} Individuals with a thermolabile variant of the enzyme methylene-tetrahydrofolate reductase can have high normal to moderately elevated levels of homocysteine.^{1,2} Homocysteine can be considered to be an independent risk factor for the development of cardiovascular disease.¹⁻³ Patients with cardiovascular disease, including heart disease, stroke, peripheral vascular disease, and thromboembolic disease generally have higher homocysteine levels than matched controls. The results of a large number of epidemiological studies have been analyzed through a meta-analysis.¹ The increased risk, or odds ratio (OR), for coronary artery disease in patients with increased homocysteine levels was estimated to be 1.7. The OR for stroke was estimated to be 2.5 and the OR for peripheral vascular disease was estimated to be 6.8. Several conditions, other than specific genetic defects or cardiovascular disease, have been associated with hyperhomocysteinemia.¹ These include vitamin deficiency, advanced age, hypothyroidism, impaired kidney function, and systemic lupus erythematosus. Medications including nicotinic acid, theophylline, methotrexate, and L-dopa have been reported to cause elevated homocysteine levels.

Specimen Requirements

Specimen

Plasma (preferred); serum is acceptable.

Volume

2 mL

Minimum Volume

1 mL

Container

Lavender-top (EDTA) tube, green-top (heparin) tube, red-top tube, or gel-barrier tube

Collection Instructions

Plasma: It is important to centrifuge blood samples for plasma collection immediately after collection to separate the plasma from the blood cells. If immediate centrifugation is not possible, collected blood specimens should be kept on ice and centrifuged within an hour.

Serum: Allow 45 minutes for serum to clot and centrifuge to separate serum from cells immediately thereafter.

Reference Range

See table.

Age	Male (μmol/L)	Female (μmol/L)
0 to 6 m	0.0–11.5	0.0–11.5
7 m to 1 y	0.0–10.0	0.0–10.0
2 to 12 y	0.0–9.0	0.0–9.0
13 to 17 y	0.0–11.0	0.0–11.0
18 to 60 y	0.0–14.5	0.0–14.5
61 to 70 y	0.0–17.2	0.0–17.2
71 to 80 y	0.0–19.2	0.0–19.2
>80 y	0.0–21.3	0.0–21.3

Storage Instructions

Room temperature. Stable at room temperature or refrigerated for 14 days. Stable frozen for six months.

Patient Preparation

A fasting specimen is preferred to establish baseline values or monitor treatment.

Causes for Rejection

Specimen received not separated from cells (do **not** respin a gel-barrier tube to harvest additional serum); gross hemolysis (eg, bright red or cherry red); whole blood tube without a gel separator; plasma from a light blue-top (sodium citrate) tube or yellow-top (ACD) tube used for coagulation studies (liquid citrate tubes have a dilutional effect of approximately 1.2 on this assay and are not approved for use); gross lipemia

References

Fortin LJ, Genest J Jr. Measurement of homocyst(e)ine in the prediction of arteriosclerosis. *Clin Biochem.* 1995; 28(2):155-162 (review). PubMed

7628074 (<http://www.ncbi.nlm.nih.gov/pubmed/7628074>)

Kang SS, Wong PW, Malinow MR. Hyperhomocyst(e)inemia as a risk factor for occlusive vascular disease. *Annu Rev Nutr.* 1992; 12:279-298. PubMed

1503807 (<http://www.ncbi.nlm.nih.gov/pubmed/1503807>)

Malinow MR. Plasma homocyst(e)ine and arterial occlusive diseases: A mini review. *Clin Chem.* 1995; 41(1):173-176 (review). PubMed 7813076

(<http://www.ncbi.nlm.nih.gov/pubmed/7813076>)

Robinson K, Mayer E, Jacobsen DW. Homocysteine and coronary artery disease. *Cleve Clin J Med.* 1994; 61(6):438-450 (review). PubMed 7828335

(<http://www.ncbi.nlm.nih.gov/pubmed/7828335>)

Ueland PM, Refsum H, Stabler SP, et al. Total homocysteine in plasma or serum: Methods and clinical applications. *Clin Chem.* 1993; 39(9):1764-1779 (review). PubMed 8375046

(<http://www.ncbi.nlm.nih.gov/pubmed/8375046>)

Verhoef P, Stampfer MJ, Buring JE, et al. Homocysteine metabolism and risk of myocardial infarction: Relation with vitamins B₆, B₁₂, and folate. *Am J Epidemiol.* 1996; 143(9):845-859. PubMed 8610698

(<http://www.ncbi.nlm.nih.gov/pubmed/8610698>)

Footnotes

1. Malinow MR, Bostom AG, Krauss RM. Homocyst(e)ine, diet, and cardiovascular diseases: A statement for healthcare professionals from the nutrition committee, American Heart Association. *Circulation.* 1999; 99(1):178-182. PubMed 9884399 (<http://www.ncbi.nlm.nih.gov/pubmed/9884399>)

2. Clarke R, Stansbie D. Assessment of homocysteine as a cardiovascular risk factor in clinical practice. *Ann Clin Biochem.* 2001; 38(Pt 6):624-632.

PubMed 11732645 (<http://www.ncbi.nlm.nih.gov/pubmed/11732645>)

3. Herrmann W. The importance of hyperhomocysteinemia as a risk factor for diseases: An overview. *Clin Chem Lab Med.* 2001; 39(8):666-674.

PubMed 11592431 (<http://www.ncbi.nlm.nih.gov/pubmed/11592431>)

